

Research Agenda: From Feasibility to Evidence

Research Agenda: From Feasibility to Evidence I

The framework identifies a large research program. [fig:research_agenda] condenses it into priorities that can be run as staged studies rather than as a single overbroad claim. The first priority is modality comparison: co-present paper PPPiP, synchronous remote DigiPPPiP, semisynchronous drawing, and asynchronous persistent-canvas practice should be compared on relational, phenomenological, behavioral, accessibility, and narrative outcomes. The comparison should include matched controls such as ordinary shared drawing, video chat, shared games, and solo art-making so DigiPPPiP is not credited for effects that any pleasant shared activity would produce.

Research Agenda: From Feasibility to Evidence II

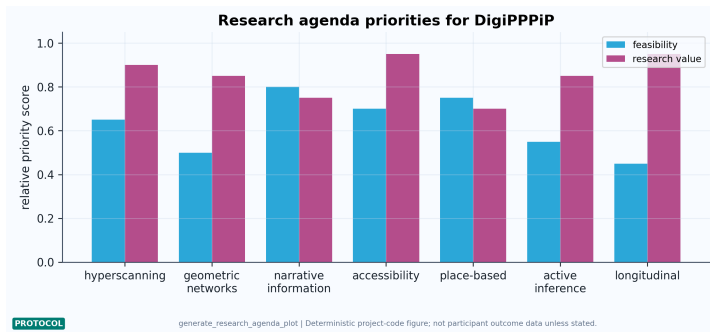


Figure 1: Conceptual research-priority map for DigiPPPiP. The figure is generated by `generate_research_agenda_plot()`; read paired bars as relative feasibility and research value assigned by the framework design for each agenda item. It supports staged prioritization rather than external funding or outcome claims. The values are conceptual planning scores, and future versions should replace them with feasibility data, stakeholder priorities, and study results.

Research Agenda: From Feasibility to Evidence III

The staged path should begin with feasibility and meaning before physiology, following the validation ladder in [fig:validation_ladder]. Early studies should ask whether partners understand the task, preserve consent, use the canvas intentionally, and describe the artifacts as relationally meaningful. Only after those conditions hold should instrumentation-heavy studies test neural, narrative, or active-inference hypotheses. This ordering follows from the design kernel: no signal-processing result can rescue a practice that participants do not experience as shared agency.

Research Agenda: From Feasibility to Evidence IV

The systems-governance agenda should be tested alongside the outcome agenda. A pilot should record whether partners use save, replay, redaction, deletion, and export controls; whether place prompts are accepted, declined, or coarsened; whether accessibility accommodations change during the session; and whether optional AI suggestions are ignored, rejected, revised, or treated as partner-like. These are not secondary usability details. They are the feedback signals that decide whether the framework's branches remain reversible and ethically bounded.

Hyperscanning studies should test whether the phase structure in [@fig:ibs_phases] appears in real dyads. Geometric-hyperscanning studies should evaluate whether the appendix's curvature proxy tracks rupture, repair, or co-regulation events after the artifact and alignment limits are addressed [@sec:formalisms_appendix]. Narrative-information studies should test whether the appendix's entropy and surprisal primitives predict partners' reports of novelty, coherence, or shared meaning [@schulz2024narrativeinfo].

Research Agenda: From Feasibility to Evidence V

Accessibility research should be participatory from the start. Disabled partners should help define input modes, feedback channels, consent controls, and successful participation. Place-based studies should test whether place-responsive prompts increase place attachment or relational memory compared with non-place prompts [lewicka2011place].

Research Agenda: From Feasibility to Evidence VI

AI-assisted studies need a separate preregistered logic.

Mixed-initiative interfaces can join automated suggestions with direct manipulation, but that does not make automation harmless or relationship-preserving [horvitz1999mixedinitiative;deterding2017mixedinitiative]. AI-mediated communication scholarship is especially relevant because it treats AI as modifying, augmenting, or generating interpersonal messages on behalf of a communicator, with consequences for agency, trust, relationship interpretation, and ethics [hancock2020aimediatedcommunication].

Companion-AI scholarship adds a distinct replacement and social-deskilling concern, so AI-supported DigiPPPiP should be measured against whether it strengthens human-human attention rather than satisfying a user through the system itself [malfacini2025companionai]. Human-AI interaction guidelines imply concrete requirements for DigiPPPiP: disclose what the model can and cannot do, let partners accept, reject, revise, or undo suggestions, support efficient dismissal, and recover gracefully from wrong or intrusive prompts [amershi2019humanai]. If the AI branch becomes a clinical or health-intervention trial, the Consolidated